Siwon Lee

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EDUCATION

Washington University in St. Louis MO, USA
Ph.D. in Economics *Aug 2024 – Present*

- Research Fields: International Trade, International Finance, Asset Pricing, Portfolio Theory, Machine Learning

Washington University in St.Louis MO, USA
M.A. in Economics *Dec 2025*

Sungkyunkwan University Seoul, Korea
B.A. in Global Economics, B.S. Mathematics *Mar 2021 – Aug 2024*

Math: Real Anal., Diff. Geom., Prob. Theory, Basic Math. for AI, Comp. Applied Math.
Software: OS, Intro DB, Data Struct., Intro AI, Sys. Prog., Algorithms
Econ: Adv. Macro, Game Theory, Econometrics, Math. Econ., Intro Fin. Acct.

LUSEM & HANKEN University Sweden and Finland
Winter Joint Exchange Program *Jan 2023*

- Visited KPMG corporation in Helsinki and discussed sustainability
- Discussed critical thinking and examples of sustainability in each country

CONFERENCE PROCEEDINGS

A Pooled Exceedance Rate Hides Two Failures in Firm Earnings-at-Risk Under Trade Policy Uncertainty
Author *ACM ICAIF 2026 proceedings (forthcoming)*

- Evaluated Earnings-at-Risk models on 402,780 firm-quarters under trade policy uncertainty
- Separated Gaussian tail misspecification from firm heterogeneity using quantile and conformal methods
- Improved coverage across firm sizes through size-specific calibration

Household Portfolio Choice under Trade Policy Uncertainty with a Deep Operator Network
Author *ACM ICAIF 2026 proceedings (forthcoming)*

- Extended Merton portfolio choice with TPU-driven jump risk and stochastic equity volatility
- Learned the two-dimensional HJB solution operator using a multiple-input DeepONet
- Enforced short-sale and adding-up constraints exactly through a softmax portfolio head
- Reduced policy evaluation from 198 ms to 0.6 ms and compared allocations with AAII survey data

WORKING PAPERS

The Impact of Tariff and Trade Policy Uncertainty Shocks on Household Asset Portfolios
Author *Working paper*

- Estimated stock, bond, and cash portfolio responses to TPU shocks using monthly local projections
- Modeled TPU shock transmission through firm inventories and imperfect inventory information
- Developed a variation of the Merton model for robust portfolio choice under Knightian uncertainty
- Characterized hedging demands against TPU shocks, ambiguity, and regime-transition risk

Domestic Firm Financial Vulnerability Consequences of Exchange Rate Depreciation
Author *Working paper*

- Estimated firm financial vulnerability after exchange rate depreciation using local projections
- Identified heterogeneous effects of trade openness and financial hedging
- Solved nonconvex equilibrium conditions using Deep Equilibrium Networks
- Developed a structural model of endogenous exchange rates and heterogeneous firms

Homogeneity, Systemic Risk, and the Regulation of Automated Trading
Author *Working paper*

- Modeled correlated automated trading, price feedback, and leveraged liquidation in continuous time
- Analyzed position limits, margin requirements, and circuit breakers with reopening auctions
- Compared regulation in India, the US, and the EU across market depth and automated trading growth

Lost in Translation or Strategic Choice? LLM Evidence on Multilingual Risk Disclosure Asymmetry
Co-author with Wonwoo Kim *Working paper*

- Compared multilingual risk disclosures of US-listed Korean and Japanese firms using LLMs
- Documented persistent cross-language differences in financial, market, and governance risk disclosure
- Developed tests to detect and mitigate cross-language measurement bias in LLM text analysis

WORK IN PROGRESS

Market Impact of Large Shareholder Stock Sales: An Agent-Based Approach
Author *Draft*

- Simulated liquidity and volatility responses to large shareholder stock sales
- Modeled heterogeneous investor behavior in limit order books using agent-based simulations
- Analyzed how information asymmetry and risk preferences amplify price impact and volatility
- Explored performative prediction feedback between investor forecasts, trading, and market prices

Labor Market Conditions and Household Financial Vulnerability
Author *Research Proposal*

- Built a state-year panel of labor market indicators and mortgage delinquency
- Estimated links between labor market vulnerability and household financial distress
- Tested lagged specifications and robustness across model variations

CONFERENCE AND WORKSHOP PRESENTATIONS

2026 ZUEL Workshop on Frontiers in Finance Research
International workshop presentation *2026*

OTHER PRESENTATION

The Effect of Internet Reviews on the Demand of Experience Goods: Case of Korean Movie Industry
Sungkyunkwan-KEIO Symposium *Jan 2024*

Stabilizing ETF Price Gaps through Liquidity Provision with AI
2022 Capstone Design and AI Hackathon *Oct 2022*

Forecasting Inflation and Economic Recession: Comparison of Econometrics and Machine Learning
2022 SKKU Annual Conference on Economics *Aug 2022*

REVIEW EXPERIENCE

2026 ICAIF Main Conference ACM
Reviewer *2026*
21st Economics Graduate Student Conference WashU, USA
Reviewer *Aug 2026*
20th Economics Graduate Student Conference WashU, USA
Reviewer *Aug 2025*

COMPLETED RESEARCH PROJECTS

Analyzing the Impact of Monetary Policy on Construction and Facility Investment SKKU, Korea
Co-author *Mar 2024 – Jul 2024*

Analyzing the Effect of Monetary Policy on Technological Progress: NAICS 334 DB, Korea
Team Leader *Dec 2023 – Feb 2024*

Comparison of South Korea's 1998 and 2008 Economic Crises SKKU, Korea
Team Member *Feb 2023 – Jun 2023*

Numerical Investigation for Economic Modeling Using Python SKKU, Korea
Team Member *Jun 2022 – Jul 2022*

Analyzing the Relationship Between COVID-19 and Trade SKKU, Korea
Team Leader *Apr 2022 – Jun 2022*

Forecasting Changing Phase of Financial Market When Implementation CBDC Korea
Co-author *Feb 2022*

Implementation of CBDC to Reduce Public Antipathy Toward Privacy Violations SKKU, Korea
Co-author *Aug 2021*

Project of Convergence Basics SKKU, Korea
Team Member *Jun 2021 – Aug 2021*

Calculated Production Possibility Frontier SKKU, Korea
Team Member *Apr 2021 – Jun 2021*

TEACHING EXPERIENCE

Assistant Instructor (WashU, USA) *Aug 2025 – Present*

- Introduction to Macroeconomics
- Introduction to Microeconomics

Teaching Assistant (SKKU, Korea) *Feb 2023 – Jun 2024*

- Algorithm and Problem Solving
- Computer Thinking and SW coding
- Problem Solving

Young Economic Education Supporters (Bank of Korea, Korea) *Apr 2023 – Dec 2023*
Lecturer

Sungkyun Tutoring (SKKU, Korea) *Feb 2022 – Jun 2024*

- Intermediate Microeconomics Theory
- Intermediate Macroeconomics Theory
- Econometrics

WORK EXPERIENCE

TRC Research Assistant UHK, Remote
Research Assistant *Nov 2023 – Jan 2024*

Wisdomagora Seoul, Korea
Editor Management Intern *Sep 2022 – Feb 2023*

EXTRACURRICULAR ACTIVITIES

Transdisciplinary Mentoring Community WashU, USA
Mentee *Aug 2024 – Jul 2025*

Global Leaders in Finance(GLIF) Seoul, Korea
Team Member *Mar 2021 – Mar 2022*

MIMIC SKKU, Korea
Team Member *Sep 2022 – Jun 2024*

Seekers SKKU, Korea
President *Mar 2022 – Jun 2024*

AWARDS

2022 SKKU Annual Conference on Economics — SKKU, Korea
Top Prize — *Aug 2022*

2022 Capstone Design and AI Hackathon — The Korean Association of Computer Education
Encouragement Prize — *Oct 2022*

Learning-Fair — SKKU, Korea
College Dept. Dean's Awards — *Nov 2021*

Sungkyun Tutoring — SKKU, Korea
Excellence Prize — *Dec 2022*

Facebook Hacker Cup 2021 — Facebook, Online
Round 2 — *Sep 2021*

Big Data Analysis Contest by NH Investment — NH Investment
Participation Prize — *Nov 2022*

Dean's List — SKKU
Aug 2022

SCHOLARSHIP

Fellowship & Full Scholarship — Aug 2024 – Aug 2029
WashU

Scholarship for Extracurricular Challenge — Jan 2024
SKKU

Scholarship for Global Learning Leader — Dec 2023
SKKU

Scholarship for Global Exchange Student — Mar 2023
SKKU

Scholarship for Academic Excellence — Aug 2022
SKKU

Scholarship for Academic Excellence — Feb 2022
SKKU

State Scholarship — Feb 2021
Government of South Korea

ADDITIONAL INFORMATION

Professional Memberships: American Finance Association
Achievements: ADsP, Certification of Cody Joint Event(MATLAB)
Skills: Programming(C, Python, MYSQL, MATLAB), LaTeX, MS Office(PowerPoint, Excel)
Languages: Korean(Native), English(Fluent)
Hobbies: Table Tennis, Tennis, Listening to Music, Basketball